



SmartPark BlockChain Project

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Group number: 12

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Project Overview – SmartPark

SmartPark Overview

- Blockchain-based hotel parking management system
- Allows users to search, book, and pay for parking spaces
- Uses smart contracts to automate bookings and payments
- Improves transparency, security, and trust between stakeholders
- Reduces reliance on a centralised parking management system

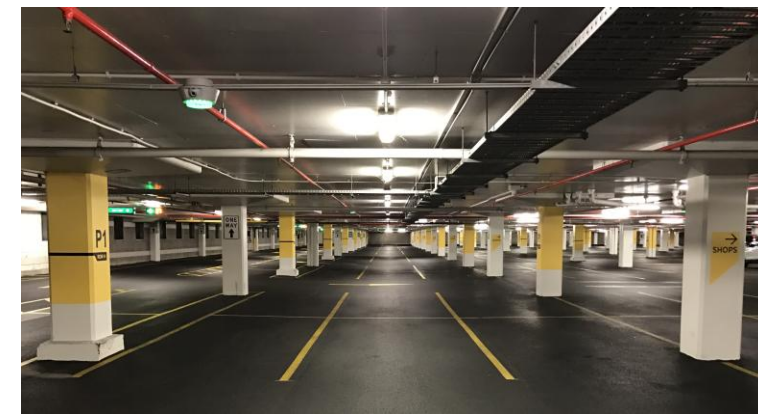
Why Use Blockchain?

- Multiple stakeholders interacting
- Shared transaction records
- Smart contract automation
- Decentralised and transparent system
- Easily trace bookings and parking spot availabilities

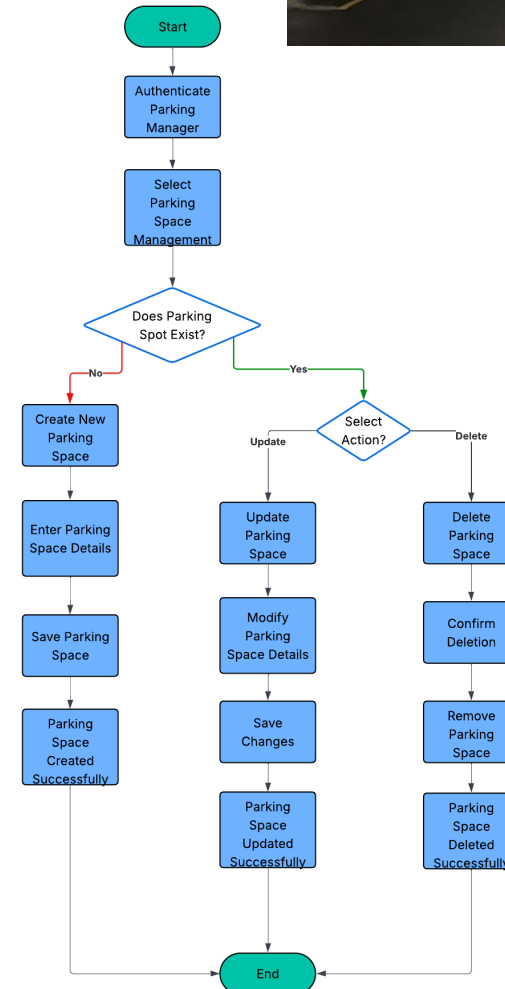
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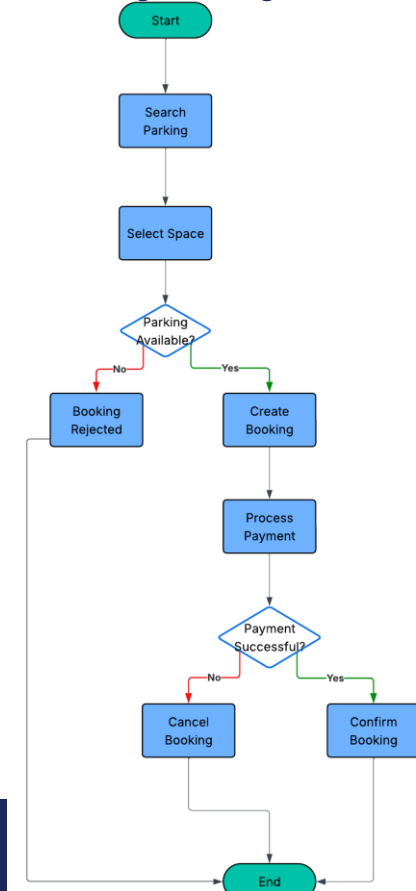
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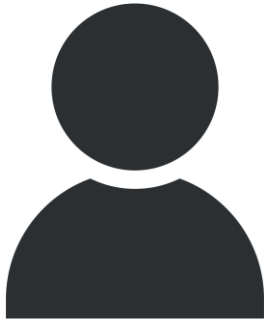
Parking User



Parking Manager



Stakeholders



- **Parking Space Provider:** Supplies, registers and manages parking spaces on the SmartPark application. Includes updating availability and pricing information.
- **Parking User:** Searches, books and pays for parking spaces through blockchain-based booking and payment transactions.
- **Blockchain System:** A decentralised blockchain application that validates, records and stores booking, parking and payment transactions securely through smart contracts.

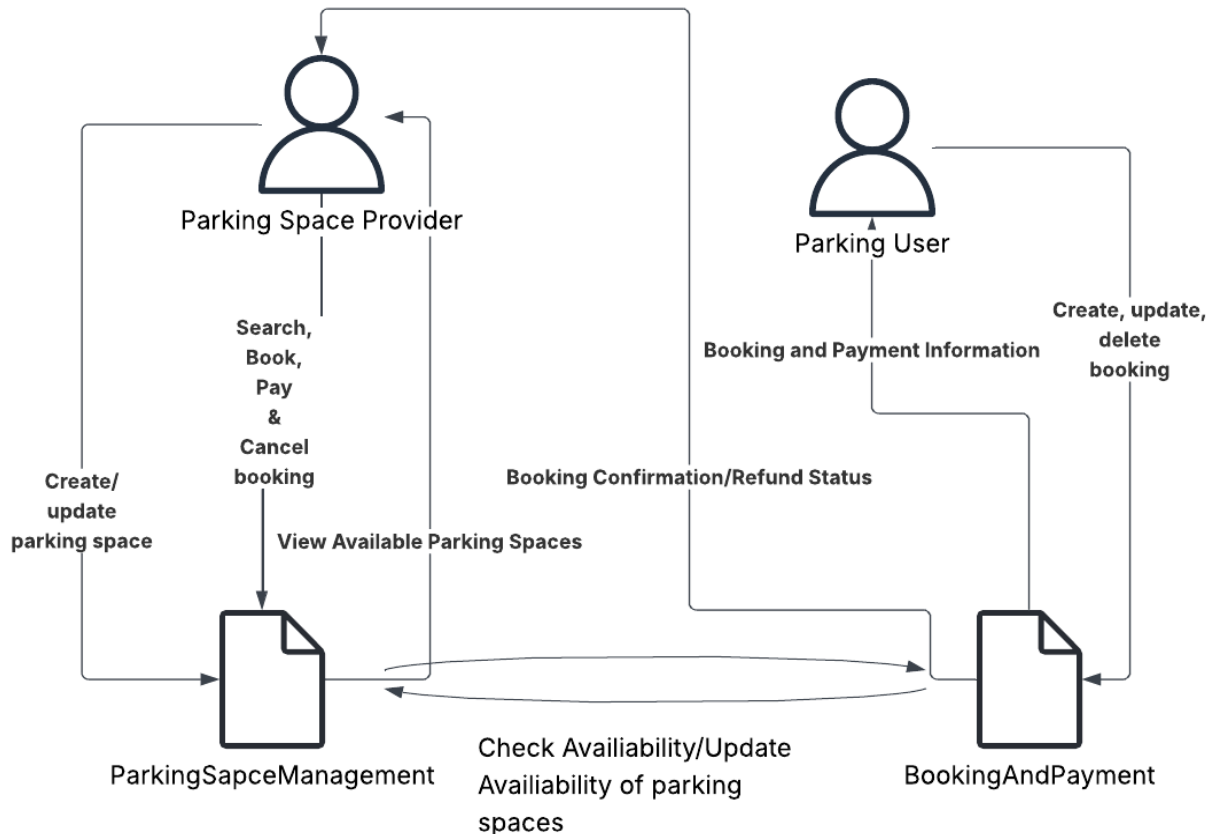
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Smart Contracts



- **ParkingSpaceManagement Contract:** Manages registration and maintenance of parking spaces. Can create, update, delete and manage parking spaces availability, pricing and location
- **BookingAndPayment Contract:** Handles the parking reservation and payment process between parking users and parking providers. Users can create, confirm, cancel, and complete bookings while securely processing payments, refunds, and payment release
- **Interaction:** The BookingAndPayment contract interacts with the ParkingSpaceManagement contract to verify parking space availability before confirming a booking.

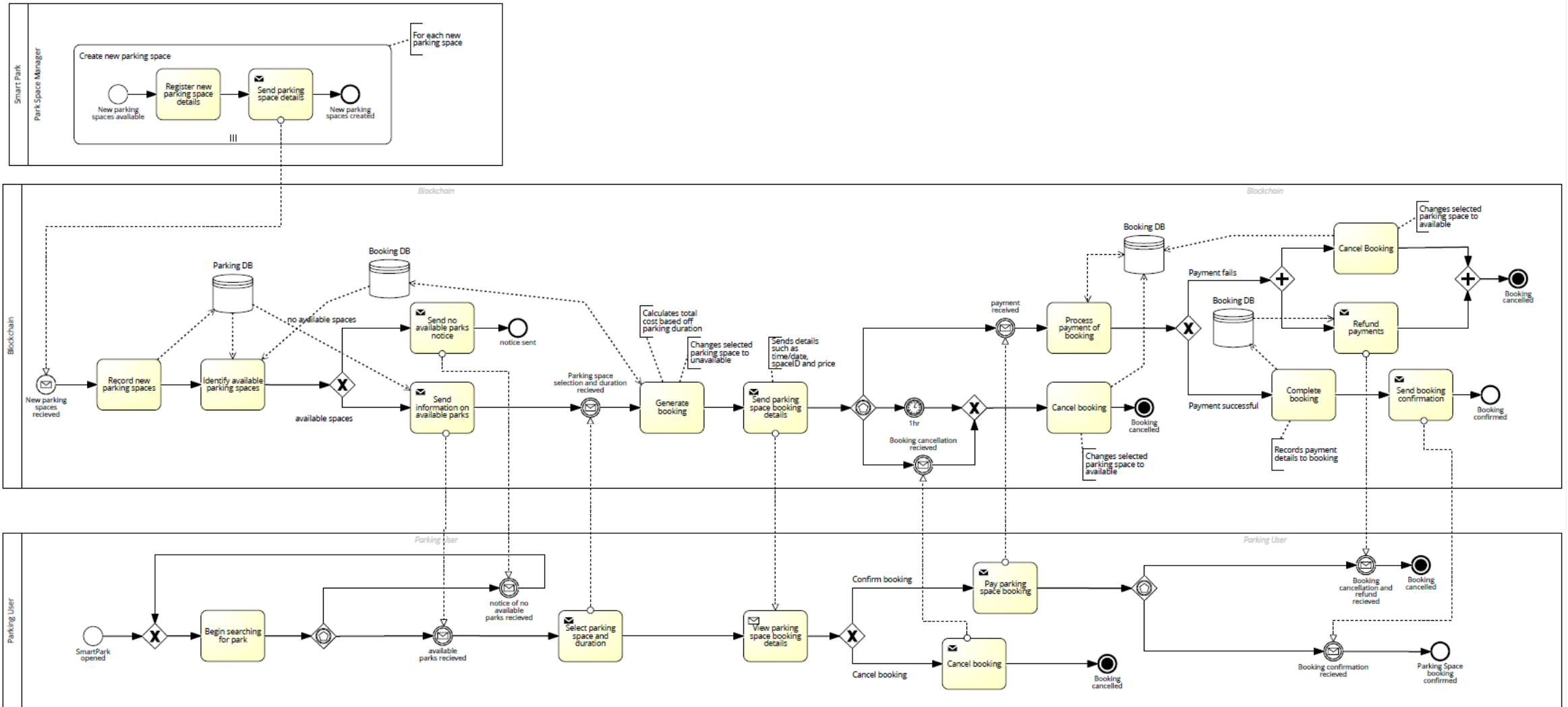
Smart Contracts Interactions



Key Interactions:

- Parking Space Provider interacts with the ParkingSpaceManagement contract to create, update, and manage parking spaces.
- Parking User interacts with the BookingAndPayment contract to create bookings, make payments and cancel bookings.
- BookingAndPayment checks parking space availability through ParkingSpaceManagement before confirming a booking.
- Once a booking is confirmed or cancelled, the availability status is updated on the blockchain.

Business Process Model Specification



TEI

Design Considerations

Back-End Considerations

- Creating parking space and booking functionality allows for two smart contracts
- Role-based access – only parking manager can perform functionalities in the ParkManagement contract
- Spaces are marked as available upon creation and given their own unique ID
- Availability of a spot is automatically updated to false whenever a bookings are made and true when bookings are cancelled
- Validation checks prevent inaccuracies around parking spaces, bookings or unauthorised updates

Front-End Considerations

- The front-end displays only available parking spaces when selecting spaces
- Simple and user-friendly interface for greater ease
- Include transaction status updates so users know whether they are pending, successful or failed



Challenges: Logic of system

- Parking space availability is a boolean showing current state as available (True) or unavailable (False)
- Prevents overlap in bookings and inaccuracies in parking availabilities

Methodology

